

ORIGINAL RESEARCH

Static and Dynamic Effects of Stroke Onset on Depressive Symptoms in Middle-Aged and Older Adults: Evidence Across 21 Countries

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BACKGROUND: Stroke and depressive symptoms are significant public health concerns worldwide; however, their causal relationship remains unclear. Therefore, there is a need for comprehensive multicountry studies that examine both static and dynamic effects over time.

METHODS AND RESULTS: We conducted a pooled multicohort study across 21 countries using nationally representative longitudinal data sets. Our sample was restricted to individuals aged ≥ 50 years who were followed for at least 2 waves. First, the fixed-effects model in conjunction with propensity score matching was used to analyze the static effect of stroke onset on depressive symptoms. Next, by using an event study methodology, we delved into the dynamic relationship. Furthermore, meta-analyses and multilevel model were conducted to assess heterogeneity and explore the moderating role of national-level indicators. Our study revealed a statistically significant and robust causal linkage between stroke onset and depressive symptoms. Further analysis highlighted both the anticipatory effects of stroke occurrence and the sustained, substantive impact on depressive symptoms over more than a decade. Meta-regression analyses indicated that countries with higher mortality from noncommunicable diseases exhibited attenuated effect sizes. Furthermore, we observed a modest moderating effect of mortality from noncommunicable diseases and life expectancy.

CONCLUSIONS: This research underscores the complex relationship between stroke and depressive symptoms, highlighting the need to further identify underlying mechanisms.

Key Words: across countries ■ depressive symptoms ■ stroke

Stroke stands as one of the predominant noncommunicable diseases (NCDs) worldwide.¹ The burden of stroke is a huge public health issue of growing importance. In 2019, stroke emerged as the second leading cause of both mortality, claiming the lives of 6.6 million individuals, and disability, contributing to 143 million disability-adjusted life-years worldwide.² During the past 3 decades, in absolute

terms, global stroke incidence increased by 70%, its prevalence increased by 85%, its mortality increased by 43%, and disability-adjusted life-years attributable to stroke have mounted by 32%, with a notably amplified burden observed in low-income and middle-income countries compared with high-income countries.² Projections derived from research indicate stroke mortality will increase by 50%, from 6.6 million

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CLINICAL PERSPECTIVE

What Is New?

- Individuals who have had a stroke often show higher levels of depressive symptoms even before stroke occurs.
- This research reveals a sustained decrease in mental well-being following a stroke, lasting up to a decade compared with those who do not experience a stroke.

What Are the Clinical Implications?

- Our study highlights the necessity of implementing routine screening measures for mental health issues within the general population; it emphasizes the critical need for accessible, responsive, and high-quality rehabilitation services for stroke survivors.

Nonstandard Abbreviations and Acronyms

CHARLS	China Health and Retirement Longitudinal Study
ELSA	English Longitudinal Study of Aging
HRS	Health and Retirement Study
KLoSA	Korean Longitudinal Study of Aging
MHAS	Mexican Health and Aging Study
NCD	noncommunicable disease
PSD	poststroke depression
SHARE	Survey of Health, Aging, and Retirement in Europe

in 2020 to 9.7 million in 2050, with disability-adjusted life-years growing over the same period, from 144.8 million in 2020 to 189.3 million in 2050.³ The burden of disability after a stroke is also large, and is increasing at a faster pace in low-income and middle-income countries than in high-income countries.³ However, it is worth noting that the complications arising from stroke often prove to be more distressing than the stroke itself.⁴ The condition is often associated with psychological impairment and the loss of behavioral and social skills.⁵

Depression is the most common psychiatric disorder in patients with stroke, characterized by a prolonged decline in emotional well-being.⁴ Poststroke depression (PSD) commonly refers to a depressive state after stroke onset, representing the most prevalent and burdensome neuropsychiatric complication following stroke.^{6,7} Especially in recent years, the

increasing number of stroke survivors has led to a significant rise in the overall prevalence of patients with PSD.⁸ Although the prevalence estimates of depression vary depending on assessment measures, diagnostic criteria, sample selection, and time of assessment, researchers have consistently reported approximately one-third of stroke survivors suffered from PSD,^{4,9,10} and the frequency is highest in the first year following stroke onset.^{11,12} Nevertheless, it is important to note that variations in research criteria, follow-up time points, and sample sizes across studies have led to inconsistencies in methodological rigor. Moreover, due to limitations in available data and study design, current research remains restricted and unable to explore the short- and long-term trajectory of PSD.

Furthermore, given the complex relationship between depression and stroke onset, there exists a paucity of scholarly focus on prestroke depression. On one hand, prestroke depression can contribute to an increased risk of stroke,¹³ with depression serving as an independent risk factor for stroke.¹⁰ On the other hand, depression before stroke can lead to an increased risk of PSD. A recent meta-analysis indicated that pooled prevalence of prestroke depression was 11.6% (95% CI, 9.2–14.7), and concluded that depression substantially amplifies the likelihood of developing PSD.^{4,7}

In summary, the existing literature is constrained by several limitations. First, many studies have focused primarily on poststroke depression and have not addressed the trajectory, overlooking the nuanced and enduring interplay relationship between depression and stroke. Second, most existing studies focus on correlational rather than causal analyses. However, stroke is a reasonably objective measure of sudden and drastic health deterioration,¹⁴ which to some extent reflects the randomization of stroke occurrence over time within the same individual, which allows us to identify the causal relationship. Last, and crucially for our investigation, prior studies suffer from shortcomings pertaining to cross-cultural generalizability, statistical conclusion validity, and measurement validity due to reliance on single data sources and inconsistent study designs and measurement methods. It is limited in fully capturing the moderating effects of national-level indicators such as health care quality and economic conditions, considering the significant role that macro-level factors play in health care delivery.

To address the research gaps, our study was designed to examine the relationship from both static and dynamic perspectives by leveraging data from 6 nationally representative longitudinal data sets spanning 21 countries and adopting a causal inference-based research framework.

METHODS

Data Availability

The data are publicly available.

Data Source

This pooled multicohort study draws upon data spanning 21 countries sourced from 6 distinct longitudinal studies: the HRS (Health and Retirement Study), the ELSA (English Longitudinal Study of Aging), the SHARE (Survey of Health, Aging, and Retirement in Europe), the KLoSA (Korean Longitudinal Study of Aging), the MHAS (Mexican Health and Aging Study), and the CHARLS (China Health and Retirement Longitudinal Study). These studies were designed using similar survey protocols to enable cross-regional comparisons. Being part of the HRS family studies, which are longitudinal panel studies surveying representative samples using a multistage, probability sampling approach, these data sets adhere to the same biennial design and standardized survey protocols. This ensures consistency in measurement of economic status, lifestyle, and health across nationally representative samples of middle-aged and older adults (aged ≥ 45 or 50 years). Detailed descriptions of each study can be found in prior publications.^{15–19} To ensure the analysis captures dynamic effects while minimizing confounding factors, such as changes in disease patterns, health care service delivery models, and individual health-seeking behaviors over extended time periods, this study excluded samples from earlier survey waves. Instead, data from waves 5 to 15 of the HRS (2000–2020), waves 1 to 9 of ELSA (2002–2018), waves 5 to 9 of SHARE (2010–2019), waves 1 to 8 of KLoSA (2006–2020), waves 3 to 5 of MHAS (2012–2018), and waves 1 to 4 of CHARLS (2011–2018) were used. All studies received ethical approval from local research ethics committees, and participants provided written informed consent before their involvement in the studies (Table S1).

Sample Construction

For consistency across all data sets, participants in this study were restricted to those aged ≥ 50 years. The sample consisted of stroke survivors who experienced a stroke during the follow-up period and individuals who remained stroke-free throughout the survey. Stroke measurement was based on whether a doctor had ever informed the respondent that they had a stroke. To construct the sample, the following approach was adopted: observations involving individuals who had experienced a stroke before entering the survey were omitted, because the primary focus was on the effects of a sudden and unforeseen health change induced by stroke. Furthermore, to construct individual-level panel data, the sample was restricted

to individuals whose depressive symptom outcomes were observed for at least 2 periods, given that the identification process depended on accounting for individual fixed effects.

To achieve the goals of causal inference, we implemented several steps for preprocessing our sample. Specifically, we used propensity score matching methods to pair individuals who experienced a stroke with similar individuals from a pool of control participants who had not experienced a stroke. This procedure effectively prunes the data set, ensuring a comparable set of treated (stroke) and control (nonstroke) individuals.

Measures

Depressive Symptoms

Given the variation in depressive symptom measurement tools across the 21 nations (Table S2), data comparability was ensured through annual standardization of depressive symptoms within each country using Z scores.

Matching Covariates

To estimate the propensity score for stroke onset, a logit model was used encompassing the following covariates: (1) demographics and socioeconomic indicators such as age, age square, sex, marital status, education attainment, work status, and household wealth; (2) health behaviors such as alcohol use, tobacco use, and physical exercise; and (3) chronic diseases such as high blood pressure, diabetes, and lung disease.

Time-Varying Covariates

Three time-varying variables were added in the analysis to improve estimation precision: age (a continuous variable), marital status (living with a partner/spouse versus not living with a partner/spouse), and number of people living in the household (a continuous variable), which directly influence an individual's mental health. To avoid bias introduced by bad controls (ie, variables that lie on the causal pathway between the treatment and the outcome), we excluded mediating variables such as unemployment and lifestyle changes, which can result from a stroke and subsequently influence depressive symptoms. Including these variables as controls would result in more biased estimates.²⁰ Therefore, no additional time-varying covariates were included in the analysis.

National Indicators

To analyze the moderating effects of national-level factors, we selected 4 indicators representing international

economic level and health care service capability: country wealth (gross domestic product per capita in US dollars), health expenditure as a proportion of gross domestic product, NCD mortality rates, and life expectancy per capita.

Statistical Analysis

Matching

To find individuals whose mental health trajectories arguably provide the right counterfactual for individuals with stroke, 1 to 2 nearest neighbor matching is implemented within propensity score calipers.²¹ To generate the propensity score, a logistic regression model is estimated using covariates measured at baseline to predict the likelihood of experiencing a stroke.

Fixed-Effects Model

After the preprocessing step (matching), we used a combination of propensity score weighting and fixed-effect regression, which ensures that the estimates remain consistent even if either the matching or the regression step, but not necessarily both, is correctly specified, a property known as double robustness.^{22,23} Specifically, we applied a fixed-effect estimator that relied solely on within-individual variation. By tracking the same individuals both before and after the occurrence of stroke, unobserved time-invariant individual characteristics can be considered, allowing for the separation of the effects of stroke onset from other confounding factors (Equation 1).

In these analyses, the year of stroke occurrence is designated as the initial year when an individual experiences a stroke.

$$y_{it} = \delta \text{Stroke_Onset}_{it} + \mathbf{X}'_{it}\alpha + (\mathbf{X}_v \times \lambda_t)' \beta + \mu_i + \gamma_t + \epsilon_{it} \quad (1)$$

where y_{it} is depressive symptoms, vector $\mathbf{X}'_{it}$ is a set of time-varying variables vector, $\mathbf{X}_v$ is a vector of baseline individual's depressive symptoms and total assets, λ_t is a time trends term, γ_t and μ_i are the year and individual fixed effects, and ϵ_{it} is the error term. The variable Stroke_Onset_{it} is a dummy variable that turns on in the year during which a person experienced stroke. The coefficient δ represents the parameter of interest, capturing the average effects of stroke across both the pre- and poststroke periods. Through the inclusion of individual fixed effects, we can control for unobserved differences across individuals as long as the differences are time-invariant. Given that stroke precipitates a sudden decline in physical health devoid of preceding effects, the timing of stroke can be deemed plausibly exogenous to the underlying trend in mental health. For all model

specifications, standard errors are clustered by individual. We conducted the analysis for each country using the fixed-effects model.

Dynamic Effects

To explore the dynamic effects, we pooled data from all countries and estimated a more flexible version of Equation 2 that incorporates stroke leads and lags. For each individual, relative time is used to identify the period of stroke occurrence: prestroke periods are indicated by negative relative values, whereas poststroke periods are indicated by positive relative values. This expanded model allows for the examination of how the effects of stroke on depressive symptoms evolve over time, capturing potential short-term and long-term impacts.

$$y_{it} = \delta_k \times \sum_{k=-9}^7 D_{it} + (\mathbf{X}_v \times \lambda_t)' \beta + \mu_i + \gamma_t + \epsilon_{it} \quad (2)$$

where D_{gt} represents a set of indicator variables that assume a value of 1 if the individual i experienced a stroke at survey wave t . Observations >9 years before suffering stroke are included in the $k = -9$ dummy, and observations >7 years after stroke onset are included in the $k = 7$ dummy. Due to the inclusion of individual fixed effects μ_i , 1 of the leads or lags must be omitted to avoid multicollinearity. We exclude the dummy variable corresponding to the year before stroke for the 2 preceding periods, denoted as $k = -2$. The coefficients of interest are δ_k , which capture the effects of stroke over the period.

To ensure the robustness of our main results, several sensitivity checks were conducted: (1) to analyze whether missing data could affect the study results, (2) to verify that our results are not driven by the choice of statistical methods, (3) and to assess the potential existence of omitted variables bias in our analysis (see Results section). All analyses were performed using Stata (version 17.0).

RESULTS

Descriptive Analysis

The mean age of respondents across different countries ranged from 63.2 to 69.74 years. Generally, there was a higher proportion of women participating in the surveys, ranging from 51.33% to 61.28%. Approximately 7 out of 10 respondents were married. Educational attainment varied significantly across countries, with Mexico, Spain, and Italy reporting approximately 10% of the population with a college degree or above, whereas China had the lowest proportion at 4.12% (Table 1).

Table 1. Basic Demographics by Country

Demographic	China	England	United States	South Korea	Mexico	Austria	Germany	Sweden	Spain	Italy	France
Sex, n (%)											
Men	26 101 (48.67)	37 542 (44.07)	80 478 (40.37)	24 887 (42.83)	19 372 (41.87)	69 48 (41.3)	81 44 (46.35)	68 71 (45.48)	99 56 (44.59)	87 49 (44.38)	81 76 (42.19)
Women	27 530 (51.33)	47 642 (55.93)	115 939 (59.03)	33 219 (57.17)	26 897 (58.13)	98 74 (58.7)	94 25 (53.65)	82 36 (54.52)	12 371 (55.41)	10 964 (55.62)	11 204 (57.81)
Missing values, n (%)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Age, $\bar{y}$, mean $\pm$ SD	63.2 $\pm$ 8.77	66.3 $\pm$ 10.28	67.08 $\pm$ 11.21	67.05 $\pm$ 10.3	64.58 $\pm$ 11.09	67.41 $\pm$ 9.98	66.06 $\pm$ 9.88	69.74 $\pm$ 9.28	69.08 $\pm$ 11	67.03 $\pm$ 10.04	67.09 $\pm$ 10.86
Missing values, n (%)	283 (0.53)	1445 (1.7)	0 (0)	584 (1.01)	79 (0.17)	347 (2.06)	98 (0.56)	104 (0.69)	247 (1.11)	186 (0.94)	166 (0.86)
Marital status, n (%)											
Not married	8299 (15.5)	27 494 (32.84)	78 618 (40.06)	13 831 (24.05)	17 931 (38.8)	62 99 (38.3)	47 68 (27.31)	49 87 (33.3)	57 96 (26.34)	46 24 (23.69)	71 86 (37.45)
Married	45 237 (84.5)	56 227 (67.16)	117 623 (59.94)	43 690 (75.95)	28 287 (61.2)	10 148 (61.7)	12 694 (72.69)	99 91 (66.7)	16 208 (73.66)	14 891 (76.31)	12 002 (62.55)
Missing values, n (%)	95 (0.18)	1463 (1.72)	161 (0.08)	585 (1.01)	51 (0.11)	375 (2.23)	107 (0.61)	129 (0.85)	323 (1.45)	198 (1)	192 (0.99)
Education attainment, n (%)											
Middle school or below	47 309 (88.21)	29 763 (38.2)	49 075 (24.99)	9130 (15.71)	38 718 (84.58)	4070 (24.19)	2048 (11.66)	5379 (35.61)	17 685 (79.21)	13 621 (69.1)	8042 (41.5)
High school	4115 (7.67)	16 232 (20.83)	57 265 (29.16)	15 863 (27.3)	1560 (3.41)	7636 (45.39)	9279 (52.81)	3511 (23.24)	1996 (8.94)	3936 (19.97)	7028 (36.26)
College or above	2207 (4.12)	31 921 (40.97)	90 040 (45.85)	33 113 (56.99)	5497 (12.01)	5116 (30.41)	6242 (35.53)	6217 (41.15)	2646 (11.85)	2156 (10.94)	4310 (22.24)
Missing values, n (%)	0 (0)	7268 (8.53)	37 (0.02)	0 (0)	494 (1.07)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Work status, n (%)											
No	21 095 (39.75)	54 183 (64.72)	119 912 (61.23)	34 717 (60.35)	28 009 (60.8)	12 327 (75.74)	10 108 (58.37)	8734 (58.89)	16 809 (76.93)	14 685 (75.52)	13 308 (70.08)
Yes	31 973 (60.25)	29 533 (35.28)	75 921 (38.77)	22 805 (39.65)	18 058 (39.2)	3949 (24.26)	7209 (41.63)	6098 (41.11)	5041 (23.07)	4759 (24.48)	5681 (29.92)
Missing values, n (%)	563 (1.05)	1468 (1.72)	578 (0.29)	584 (1.01)	202 (0.44)	546 (3.25)	252 (1.43)	275 (1.82)	477 (2.14)	269 (1.36)	391 (2.02)
People no., mean $\pm$ SD	3.2 $\pm$ 1.7	2.05 $\pm$ 0.9	2.28 $\pm$ 1.25	2.67 $\pm$ 1.27	2.61 $\pm$ 1.89	1.96 $\pm$ 0.93	2.03 $\pm$ 0.77	1.83 $\pm$ 0.6	2.35 $\pm$ 1.03	2.42 $\pm$ 1.05	1.97 $\pm$ 0.89
Missing values, n (%)	88 (0.16)	1445 (1.7)	0 (0)	584 (1.01)	51 (0.11)	347 (2.06)	98 (0.56)	103 (0.68)	247 (1.11)	182 (0.92)	166 (0.86)
Alcohol use, n (%)											
No	29 645 (55.51)	8743 (12.08)	92 709 (47.22)	3653 (6.35)	34 731 (75.18)	6388 (46.07)	6300 (43.1)	5101 (39.34)	12 004 (64.1)	9713 (58.44)	6510 (38.15)
Yes	23 764 (44.49)	63 623 (87.92)	103 621 (52.78)	53 869 (93.65)	11 469 (24.82)	7478 (53.93)	8318 (56.9)	7867 (60.66)	6724 (35.9)	6907 (41.56)	10 556 (61.85)
Missing values, n (%)	222 (0.41)	12 818 (15.05)	74 (0.04)	584 (1.01)	69 (0.15)	2956 (17.57)	2951 (16.8)	2139 (14.16)	3599 (16.12)	3093 (15.69)	2314 (11.94)

(Continued)

Table 1. Continued

Demographic	China	England	United States	South Korea	Mexico	Austria	Germany	Sweden	Spain	Italy	France
Tobacco use, n (%)											
No	36866 (73.63)	72305 (86.96)	168 148 (86.11)	49412 (85.9)	40725 (88.19)	8634 (80.9)	8413 (81.8)	8019 (88.69)	11 507 (87)	10 423 (84.1)	11 294 (84.53)
Yes	13206 (26.37)	10841 (13.04)	27 123 (13.89)	8109 (14.1)	5455 (11.81)	2039 (19.1)	1872 (18.2)	1023 (11.31)	1720 (13)	1970 (15.9)	2067 (15.47)
Missing values, n (%)	3559 (6.64)	2038 (2.39)	1127 (0.57)	585 (1.01)	89 (0.19)	2954 (17.56)	2949 (16.79)	2135 (14.13)	3595 (16.1)	3094 (15.7)	2310 (11.92)
Physical exercise, n (%)											
No	14678 (49.35)	15266 (18.28)	36412 (22.38)	37 140 (64.57)	26706 (62.25)	1388 (10.01)	1197 (8.19)	725 (5.59)	3564 (19.03)	3891 (23.41)	2139 (12.54)
Yes	15063 (50.65)	68241 (81.72)	126307 (77.62)	20382 (35.43)	16 192 (37.75)	12 480 (89.99)	13 421 (91.81)	12 243 (94.41)	15 165 (80.97)	12 729 (76.59)	14 925 (87.46)
Missing values, n (%)	23890 (44.55)	1677 (1.97)	33686 (17.15)	584 (1.01)	3371 (7.29)	2954 (17.56)	2951 (16.8)	2139 (14.16)	3598 (16.12)	3093 (15.69)	2316 (11.95)
Stroke, n (%)											
No	52 198 (97.33)	81651 (95.85)	187 326 (95.37)	55884 (96.18)	45 726 (98.83)	15665 (93.12)	17 116 (97.42)	14 714 (97.4)	21 604 (96.76)	19 176 (97.28)	18 740 (96.7)
Yes	1433 (2.67)	3533 (4.15)	9091 (4.63)	2222 (3.82)	543 (1.17)	1157 (6.88)	453 (2.58)	393 (2.6)	723 (3.24)	537 (2.72)	640 (3.3)
Missing values, n (%)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Depressive symptoms, mean±SD	8.3 ±6.29	1.43 ±1.9	1.49 ±1.99	6.74 ±5.38	3.29 ±2.61	1.96 ±1.97	2.26 ±1.99	1.95 ±1.82	2.64 ±2.64	2.82 ±2.54	2.83 ±2.27
Missing values, n (%)	4172 (7.78)	5237 (6.15)	12 278 (6.25)	605 (1.04)	3359 (7.26)	2906 (17.27)	2936 (16.71)	2118 (14.02)	3575 (16.01)	3085 (15.65)	2270 (11.71)
Demographic	Denmark	Greece	Switzerland	Belgium	Israel	Czech Republic	Poland	Luxembourg	Slovenia	Estonia	
Sex, n (%)											
Men	6798 (45.68)	4489 (42.41)	6093 (44.83)	10158 (44.53)	2969 (42)	8888 (40.34)	4251 (43.55)	2406 (45.85)	6597 (42.2)	9697 (38.72)	
Women	8083 (54.32)	6096 (57.59)	7498 (55.17)	12 654 (55.47)	4100 (58)	13 143 (59.66)	5510 (56.45)	2841 (54.15)	9035 (57.8)	15348 (61.28)	
Missing values, n (%)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	
Age, y, mean±SD	65.44±10.21	67.76±10.09	67.16±10.15	65.83±10.78	68.79±9.94	67.21±9.35	65.99±9.77	65.36±9.58	67.38±9.94	67.6±10.36	
Missing values, n (%)	87 (0.58)	18 (0.17)	50 (0.37)	252 (1.1)	85 (1.2)	318 (1.44)	40 (0.41)	30 (0.57)	144 (0.92)	364 (1.45)	
Marital status, n (%)											
Not married	4613 (31.18)	2813 (26.62)	4460 (32.95)	8359 (37.08)	1928 (27.77)	7896 (36.4)	2829 (29.12)	1286 (24.65)	4634 (29.93)	10 298 (41.74)	
Married	10 181 (68.82)	7755 (73.38)	9074 (67.05)	14 187 (62.92)	5015 (72.23)	13 799 (63.6)	6887 (70.88)	3931 (75.35)	10 850 (70.07)	14 376 (58.26)	
Missing values, n (%)	87 (0.58)	17 (0.16)	57 (0.42)	266 (1.17)	126 (1.78)	336 (1.53)	45 (0.46)	30 (0.57)	148 (0.95)	371 (1.48)	
Education attainment, n (%)											
Middle school or below	2578 (17.32)	5412 (51.13)	2634 (19.38)	8852 (38.8)	2349 (33.23)	8877 (40.29)	3146 (32.23)	2285 (43.55)	5202 (33.28)	6681 (26.68)	

(Continued)

Table 1. Continued

Demographic	Denmark	Greece	Switzerland	Belgium	Israel	Czech Republic	Poland	Luxembourg	Slovenia	Estonia
High school	5748 (38.63)	2804 (26.49)	6385 (46.98)	6083 (26.67)	1786 (25.27)	9643 (43.77)	5099 (52.24)	1815 (34.59)	7340 (46.95)	8439 (33.7)
College or above	6555 (44.05)	2369 (22.38)	4572 (33.64)	7877 (34.53)	2934 (41.51)	3511 (15.94)	1516 (15.53)	1147 (21.86)	3090 (19.77)	9925 (39.63)
Missing values, n (%)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Work status, n (%)										
No	7299 (49.87)	8007 (75.89)	7248 (54.23)	14862 (67.32)	4238 (61.64)	15000 (69.96)	7092 (73.14)	3809 (74.35)	12660 (82.24)	15234 (62.09)
Yes	7336 (50.13)	2544 (24.11)	6118 (45.77)	7214 (32.68)	2637 (38.36)	6440 (30.04)	2604 (26.86)	1314 (25.65)	2734 (17.76)	9303 (37.91)
Missing values, n (%)	246 (1.65)	34 (0.32)	225 (1.66)	736 (3.23)	194 (2.74)	591 (2.68)	65 (0.67)	124 (2.36)	238 (1.52)	508 (2.03)
People no., mean±SD	1.96±0.76	2.21±0.96	2.01±0.84	2.06±0.95	2.25±1.1	2.06±0.98	2.67±1.49	2.32±1.1	2.33±1.14	1.99±0.91
Missing values, n (%)	87 (0.58)	17 (0.16)	50 (0.37)	252 (1.1)	84 (1.19)	317 (1.44)	40 (0.41)	29 (0.55)	144 (0.92)	364 (1.45)
Alcohol use, n (%)										
No	3282 (25.46)	4863 (51.5)	3710 (31.17)	6897 (35.84)	4262 (84.78)	10325 (55.72)	4785 (75.08)	1414 (35.33)	6852 (57.39)	15079 (76.02)
Yes	9609 (74.54)	4580 (48.5)	8193 (68.83)	12348 (64.16)	765 (15.22)	8206 (44.28)	1588 (24.92)	2588 (64.67)	5088 (42.61)	4757 (23.98)
Missing values, n (%)	1990 (13.37)	1142 (10.79)	1688 (12.42)	3567 (15.64)	2042 (28.89)	3500 (15.89)	3388 (34.71)	1245 (23.73)	3692 (23.62)	5209 (20.8)
Tobacco use, n (%)										
No	7337 (80.27)	5365 (76.33)	7265 (80.75)	11763 (82.03)	2856 (86.76)	10787 (77.83)	3615 (76.46)	2500 (85.01)	8070 (85.95)	12292 (80.57)
Yes	1803 (19.73)	1664 (23.67)	1732 (19.25)	2577 (17.97)	436 (13.24)	3073 (22.17)	1113 (23.54)	441 (14.99)	1319 (14.05)	2964 (19.43)
Missing values, n (%)	1988 (13.36)	1144 (10.81)	1684 (12.39)	3563 (15.62)	2040 (28.86)	3499 (15.88)	3386 (34.69)	1237 (23.58)	3689 (23.6)	5183 (20.69)
Physical exercise, n (%)										
No	795 (6.17)	860 (9.11)	815 (6.85)	2475 (12.86)	1238 (24.71)	2002 (10.8)	1241 (19.48)	369 (9.2)	1288 (10.79)	2468 (12.43)
Yes	12093 (93.83)	8584 (90.89)	11088 (93.15)	16769 (87.14)	3773 (75.29)	16531 (89.2)	5131 (80.52)	3641 (90.8)	10651 (89.21)	17381 (87.57)
Missing values, n (%)	1993 (13.39)	1141 (10.78)	1688 (12.42)	3568 (15.64)	2058 (29.11)	3498 (15.88)	3389 (34.72)	1237 (23.58)	3693 (23.62)	5196 (20.75)
Stroke, n (%)										
No	14472 (97.25)	10448 (98.71)	13324 (98.04)	22004 (96.46)	6798 (96.17)	20978 (95.22)	9543 (97.77)	5142 (98)	15016 (96.06)	23772 (94.92)
Yes	409 (2.75)	137 (1.29)	267 (1.96)	808 (3.54)	271 (3.83)	1053 (4.78)	218 (2.23)	105 (2)	616 (3.94)	1273 (5.08)
Missing values, n (%)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Depressive symptoms, mean±SD	1.77±1.85	2.29±2.53	1.9±1.79	2.53±2.22	2.38±2.46	2.22±2.14	3.12±2.44	2.48±2.17	2.31±2.05	2.96±2.23
Missing values, n (%)	1971 (13.25)	1140 (10.77)	1655 (12.18)	3463 (15.18)	2032 (28.75)	3462 (15.71)	3385 (34.68)	1234 (23.52)	3682 (23.55)	5166 (20.63)

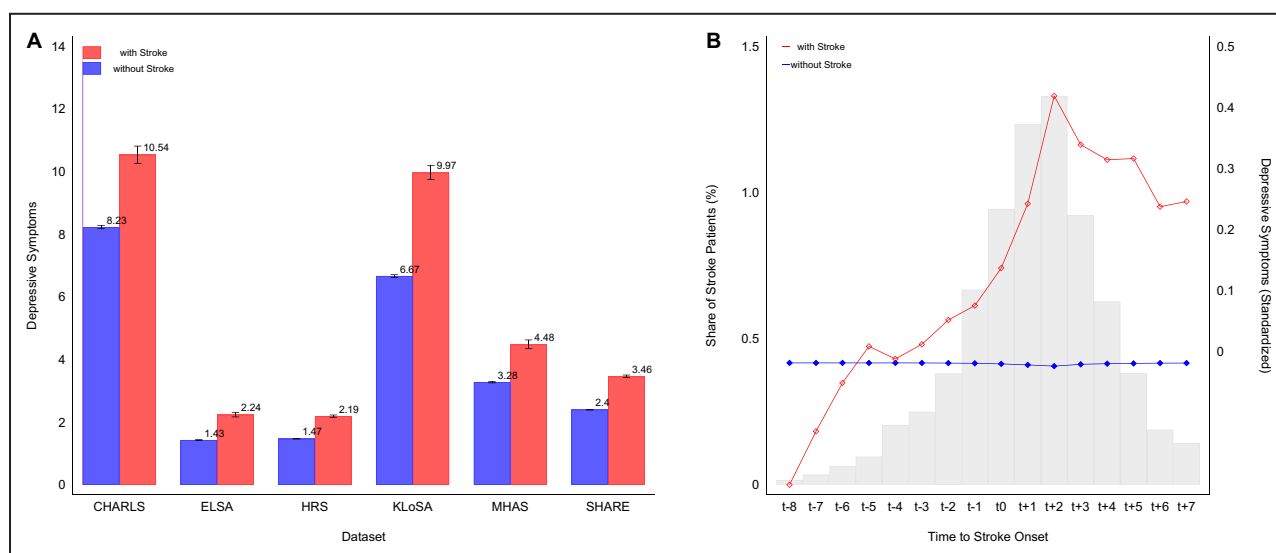


Figure 1. Differences of depressive symptoms (by stroke).

A, Difference in depressive symptom scores between individuals with and without stroke. **B**, Differences in depressive symptoms between the stroke cohort and the nonstroke cohort at different relative time periods. Negative time values represent the waves before stroke onset, whereas positive values correspond to the waves after stroke onset. The shaded bars depict the proportion of patients with stroke at each relative time period. Both parts present raw data.

In regard to stroke, the self-reported prevalence ranged from 1.17% to 6.88%. We present the original depressive symptoms scores. Due to variations in the measurement tools for depressive symptoms across different databases, the scores varied. From a cross-sectional perspective, individuals with stroke tended to have higher depression symptoms (Figure 1A).

To further analyze the changes in depressive symptoms before and after stroke onset, Figure 1B plots trends in depressive symptom changes for stroke survivors (presented as the red lines) and adults with no history of stroke (presented as the blue lines).

These data suggest that among individuals who have experienced stroke, depressive symptoms undergo significant changes before and after the event. Compared with those who have not experienced stroke, individuals who have had a stroke exhibit higher levels of depressive symptoms during the prestroke period. Due to the smaller number of individuals with stroke, especially in the distant periods from stroke onset, there are larger fluctuations in the data.

Static Effects

Figure 2 presents the coefficient estimates of Equation (1) for each country. To examine whether the inclusion of control variables affects the coefficient estimates, Figure 2 displays the results with individual and year fixed effects, added individual controls, and added all controls.

Across all countries, the associations between stroke onset and depressive symptoms are positive. Moreover, the coefficients are statistically significant in

most countries, indicating that stroke onset increases the levels of depressive symptoms (Table S3).

Dynamic Effects

Figure 3 displays the estimated leads and lags coefficients for the findings. The plot illustrates a notable increase in levels of depressive symptoms 2 years before stroke onset (t-1), followed by a significant surge in depressive symptoms in the year of onset, which continues to escalate in subsequent years. This observation lends strong support to the anticipation hypothesis, suggesting that individuals experience a significant and progressive rise in depressive symptom scores 2 to 3 years preceding stroke onset.

The results further indicate a substantial immediate (1 period) response, albeit with diminishing effects observed over 10 years following stroke.

The analysis of dynamic effects underscores that the adverse effects on mental health are both immediate and enduring, with limited evidence of long-term recovery.

Overall, the results suggest that depressive symptoms increase between 0.1 and 0.3 SDs as a result of stroke onset.

Meta-Analysis and Country Heterogeneity

One of the most distinctive aspects of our study data set is its harmonization across all 21 countries, allowing us to examine how stroke onset affects depressive symptoms differently across countries.

Subsequently, we synthesized our findings through multinational meta-analyses. Our analysis revealed a

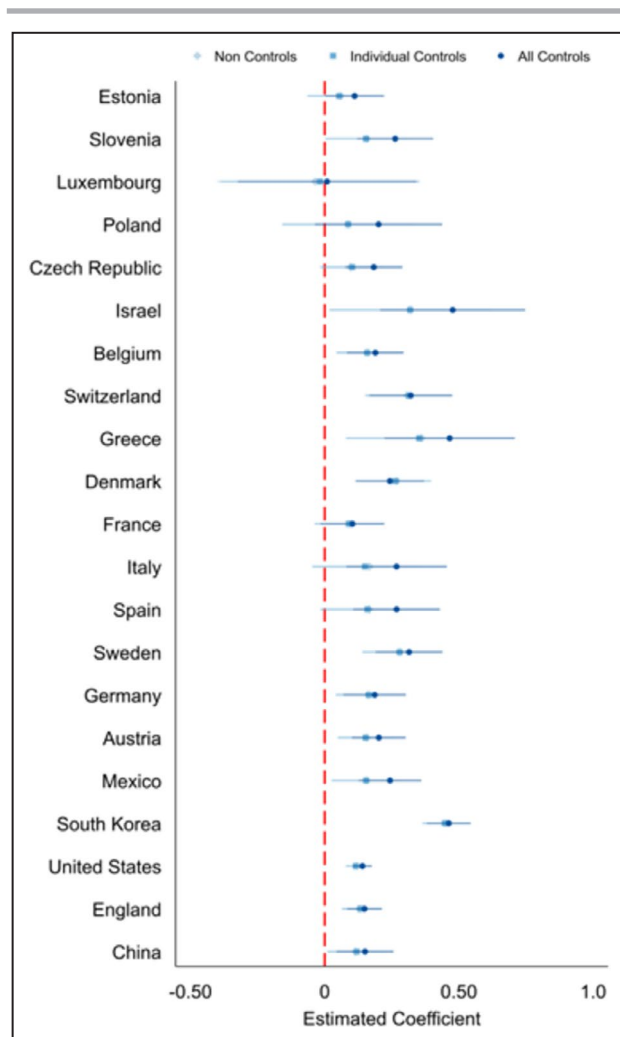


Figure 2. Static relationship between stroke onset and depressive symptoms.

Noncontrols: Only control for individual and year fixed effects. Individual controls: In addition to the above, control for time-varying variables such as age, marital status, and number of people living in the household. All controls: Further control for baseline individual characteristics (such as depressive symptoms and total household assets) interacted with year trends. The standard errors were clustered by individual. All regressions control for individual and year fixed effects.

positive association between stroke onset and depressive symptoms (pooled coefficient=0.23 [95% CI, 0.18–0.27]; I^2 =71.61%; H^2 =3.52). Here, the I^2 statistic represents the percentage of variability in effect size attributable to between-study heterogeneity, rather than sampling error. It provides insight into the degree of inconsistency among study findings. On the other hand, the H^2 statistic describes the ratio of observed variation to the expected variance due to sampling error (Figure 4).

Macro-Level Indicators

To evaluate the extent of heterogeneity in the relationship explained by country-level factors, meta-regressions

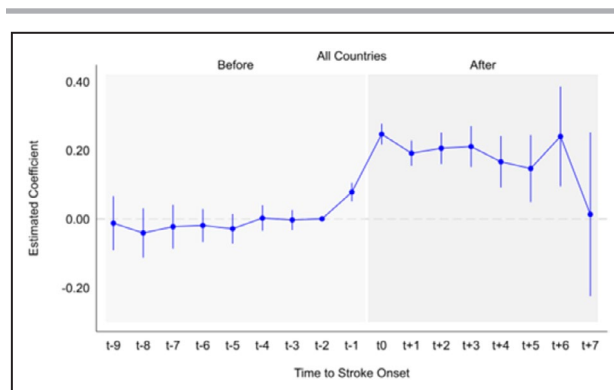


Figure 3. Dynamic effects of stroke on depressive symptoms.

The model controlled for time-varying variables including age, marital status, and number of people living in the household, and baseline individuals' characteristics (depressive symptoms, total household assets) interacted with the year trends. The standard errors were clustered by individual. All regressions control for individual and year fixed effects. Each blue dot is an estimated coefficient of a dummy variable indicating the year(s) since stroke onset (or before having a stroke). Negative time values represent the period (wave) before stroke onset, whereas positive values indicate the period (wave) after stroke.

were conducted using these factors as potential predictors of between-study heterogeneity in outcomes. Specifically, for health expenditure proportion and country wealth (Figure 5), no significant differences in effect sizes were observed based on these predictors. However, for mortality from NCDs, negative associations were identified between effect sizes and this indicator. These findings suggest that variations in NCD mortality rates may contribute to differences in the effect of stroke on depressive symptoms across countries. Specifically, for a 1-SD increase in mortality from NCDs across countries, the association between stroke onset and depressive symptoms was 0.06 and 0.044 points smaller, respectively (mortality from NCDs: coefficient=−0.060 [95% CI, 0.01–0.02]).

Furthermore, we explored whether these country-level factors could moderate the relationship between stroke onset and depressive symptoms. Through interaction analyses of stroke onset with country-level factors in multilevel models, we observed a modest moderating effect of mortality from NCDs and life expectancy on the association between stroke onset and depression. These findings suggest that individuals residing in contexts characterized by higher mortality from NCDs and lower life expectancy may experience a diminished impact of stroke onset on their mental health. Visual representations of these associations are depicted in Figure 6.

Heterogenous Analysis

To shed light on potential heterogeneous effects of stroke onset on depressive symptoms by sex, we conducted

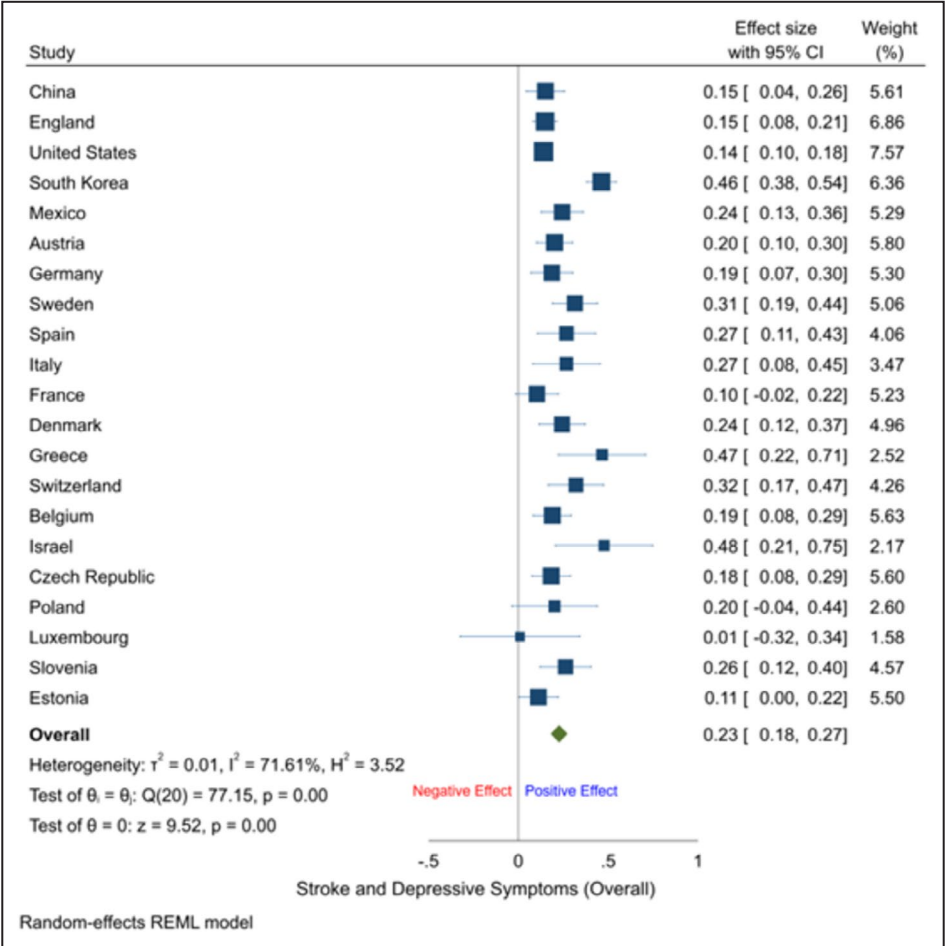


Figure 4. Meta-analysis of the findings from the static associations (n=21). Data were first analyzed separately for each country using fixed-effects regression. The findings were then pooled into multinational meta-analyses using the random-effects model to estimate the overall effect sizes for all outcomes. Between-study heterogeneity was estimated using the algorithm of the restricted maximum likelihood and was assessed using I^2 and H^2 statistics. I^2 is the percentage of variability in the effect size that is caused by between-study heterogeneity, rather than by sampling error. The H^2 statistic describes the ratio of the observed variation and the expected variance due to sampling error. Given that some of the analyses had more participants than others, and thus had lower sampling variability and more precise estimates, the meta-analysis was weighted. Studies with a greater number of respondents were given more weight than studies with a small number of respondents. These were relative weights that summed to 100. Data are presented as fixed-effects coefficients and 95% CIs. The overall effect size and its width should have accounted for the between-study variance, the number of studies, the precision of the study-specific estimates (or effect sizes), and the significance level.

the same analysis grouped by sex. [Figure S1](#) presents meta-analysis findings stratified by sex, whereas [Figure S2](#) illustrates sex-stratified dynamic effects.

The findings from the subgroup analysis are consistent with the overall results, suggesting that any insignificant heterogeneity may imply robust negative effects of stroke on mental health.

Robustness Checks

First, we reran the analysis after using a multiple imputation procedure (n=20) to impute missing data. We found that the results were largely replicated ([Table S4](#)).

Second, we refrained from using matching techniques to select samples and instead directly used a 2-way fixed-effects model to reanalyze the relationship. We found that our results were robust ([Table S5](#) and [Figure S3](#)).

Last, despite our efforts to minimize the potential presence of omitted variable bias through careful selection of identification strategies and control variables, we acknowledge that, given the nonexperimental nature of our study, there may still exist unobservable factors that could introduce bias.

To address this concern, we appeal to Oster’s method of coefficient stability as a test for selection on

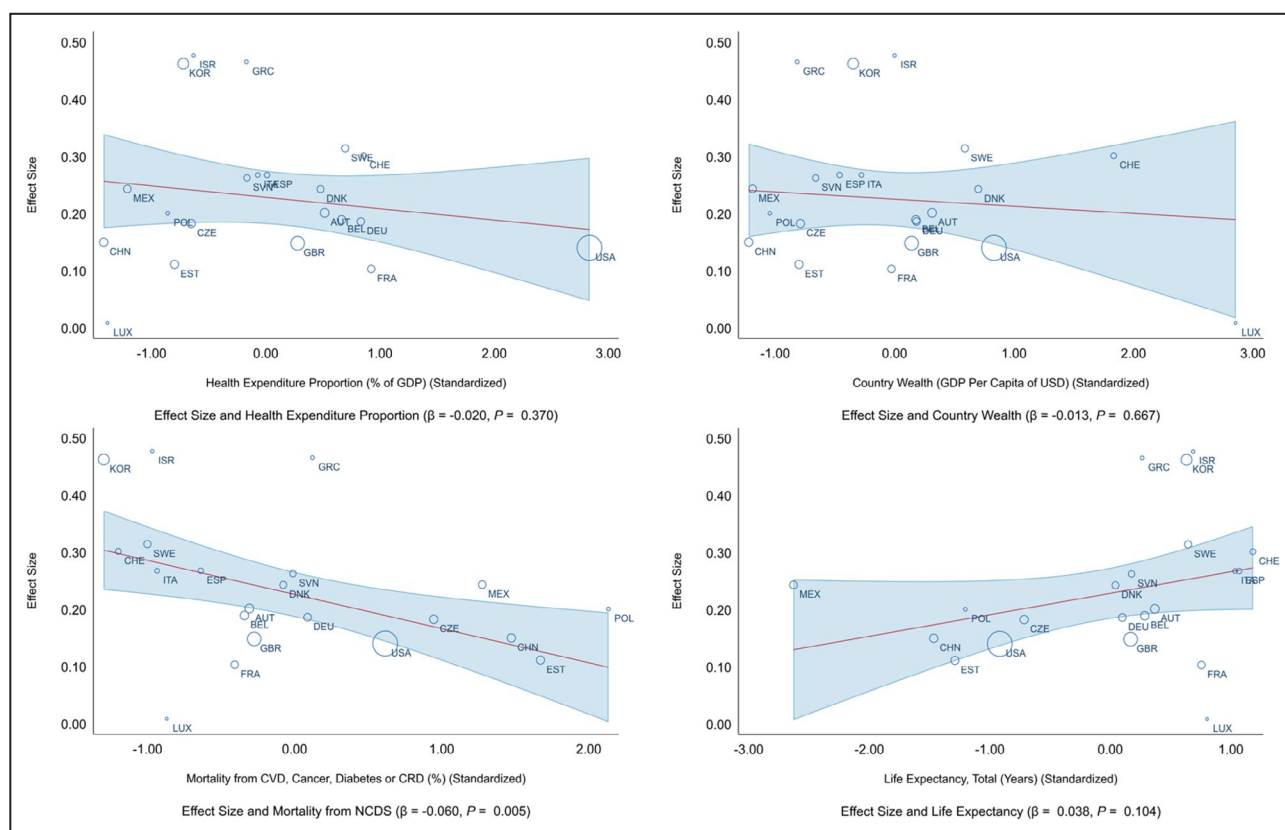


Figure 5. Bubble plot with fitted meta-regression line of the effect size (n=21).

Data are presented as fixed-effects coefficients (bubbles), 95% CIs (shaded area), and the linear prediction (red line). AUT indicates Austria; BEL, Belgium; CHE, Switzerland; CHN, China; CRD, Chronic Respiratory Diseases; CVD, cardiovascular disease; CZE, Czech Republic; DEU, Germany; DNK, Denmark; ESP, Spain; EST, Estonia; FRA, France; GBR, England; GDP, gross domestic product; GRC, Greece; ISR, Israel; ITA, Italy; KOR, South Korea; LUX, Luxembourg; MEX, Mexico; NCDS, noncommunicable diseases; POL, Poland; SVN, Slovenia; SWE, Sweden; and USA, United States.

unobservables.²⁴ The test considers both the stability of the estimated stroke onset effect when adding key observables and the importance of these factors in explaining depressive symptoms.

The estimate of the coefficients of proportionality proposed by Oster, which serves as a summary measure of the robustness of results, is calculated to be 1.862. This value exceeds the lower bound of 1, which Oster suggests as a standard for robustness (Table S6). This value implies that unobservables would need to be 86.2% more important than the observables in explaining the stroke onset treatment for the treatment effect of stroke to be 0, which seems unlikely given the richness of our data. Therefore, based on Oster's coefficient stability test, we are more confident that our finding of a significant effect of stroke onset on depressive symptoms is not merely an artifact of omitted variable bias.

This test evaluates both the stability of the estimated effect of stroke onset when additional key observables are included and the importance of these factors in explaining depressive symptoms.

DISCUSSION

This comprehensive, cross-cultural, prospective study contributes novel evidence indicating a robust link between stroke onset and depressive symptoms. Our findings suggest that individuals who experience stroke exhibit elevated depressive symptoms even before the onset of stroke. Moreover, we observe a sustained decline in mental health following stroke onset, with the effects lasting up to 10 years.

Although prior research has highlighted similar associations, our study contributes to the existing literature by examining both static and dynamic effects rigorously. We used a harmonized multicountry longitudinal framework that enables us to use a long follow-up period and alleviates endogeneity concerns. Additionally, we investigated heterogeneity in the stroke onset-depressive symptoms relationship across multiple countries, identifying a notably stronger relationship in countries with higher mortality rates from NCDs. Finally, we provided evidence in favor of a

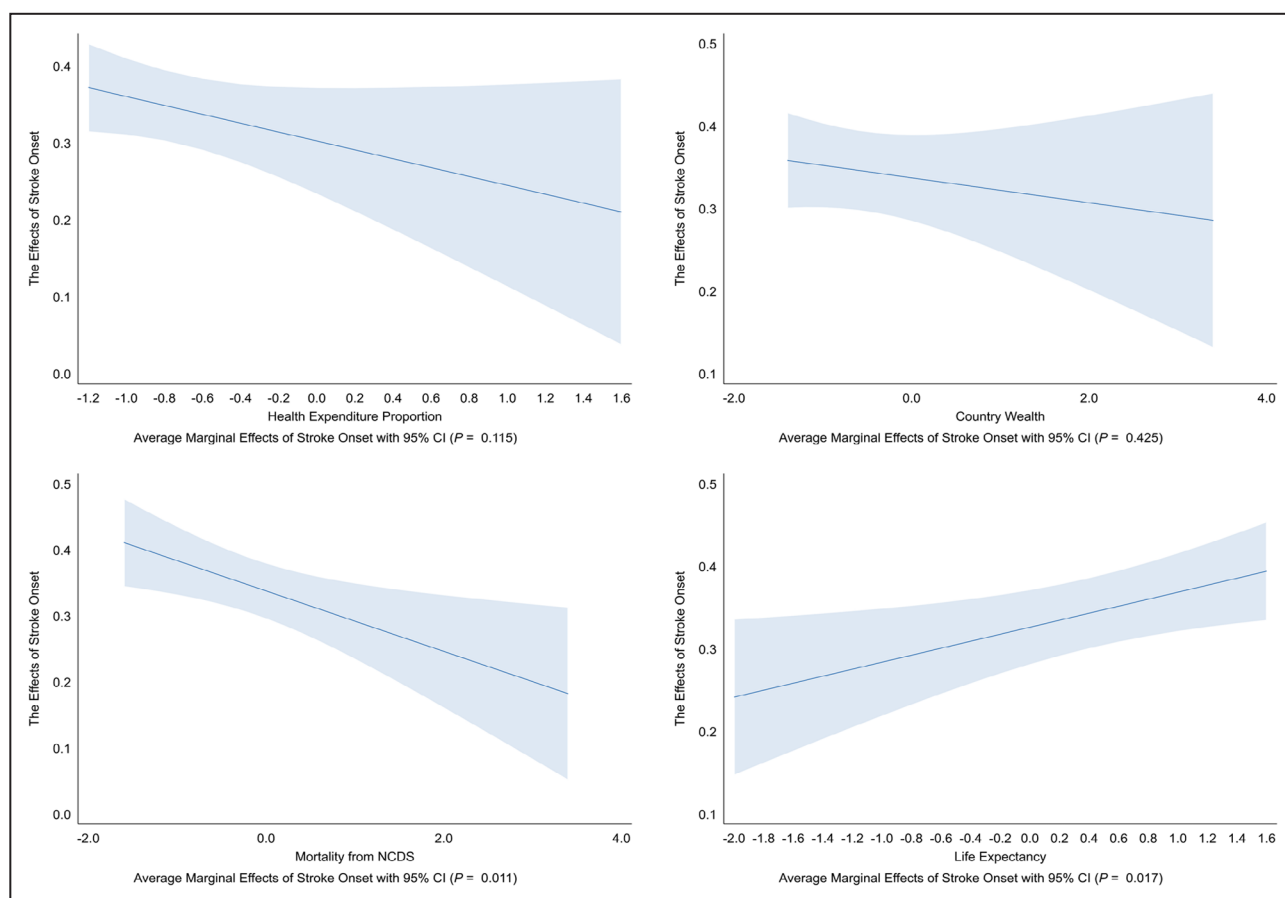


Figure 6. Moderating effects of country-level factors.

AUT indicates Austria; BEL, Belgium; CHE, Switzerland; CHN, China; CZE, Czech Republic; DEU, Germany; DNK, Denmark; ESP, Spain; EST, Estonia; FRA, France; GBR, England; GRC, Greece; ISR, Israel; ITA, Italy; KOR, South Korea; LUX, Luxembourg; MEX, Mexico; NCDs, noncommunicable diseases; POL, Poland; SVN, Slovenia; SWE, Sweden; and USA, United States.

causal interpretation of our estimates using Oster's test for selection on unobservables.

Our article increased the empirical evidence and shed further light on the changes of depressive symptoms before, during, and after the onset of stroke. From a static perspective, our findings align with existing research on PSD.^{4,25} In comparison with these studies, our research provides stronger evidence due to its consistent and rigorous research design. From a dynamic perspective, our study reveals the presence of both anticipation effects and enduring impacts of stroke onset. Namely, there is strong evidence that depressive symptoms anticipate future stroke onset occurring. This finding is consistent with results from previous research. A meta-analysis of 28 prospective cohort studies found that depression was associated with a 1.45-times increased risk of total stroke.¹³ Additionally, compared with participants with persistently low depressive symptoms, those with persistently high depressive symptoms over 2 consecutive annual assessments faced an elevated risk of stroke.²⁶ Furthermore, we observed that the negative impact on

mental health persists for ≥ 10 years after the onset of stroke, with this effect being most pronounced in the year of stroke onset. The pathogenesis of PSD mainly includes 2 theories: the primary endogenous mechanism theory and the reactive mechanism theory.²⁷ The former suggests that PSD may be linked to neurobiological changes in the brain following brain tissue damage. Damage to norepinephrine or 5-hydroxytryptamine projection fibers after stroke may result in an imbalance or dysfunction of these neurotransmitters in the brain, leading to depression.¹⁰ The latter believes that PSD is a reactive depression triggered by physiological and psychological imbalances after stroke, influenced by various factors such as family, society, genetics, and living environment. Currently, an increasing number of scholars advocate for the belief that the pathogenesis of PSD involves the joint involvement of neurobiological and sociopsychological mechanisms.^{28,29} Overall, this result reflects the severe and persistent negative impact of stroke on patients' mental health.

In regions with higher mortality from NCDs and lower life expectancy per capita, the observed impact

of stroke on depressive symptoms appears to be less pronounced. Meta-regression analyses indicated a negative correlation between mortality from NCDs and effect sizes, suggesting that the effect sizes of studies increased in line with the mortality rate due to chronic diseases at the country level. Moreover, the multiple level model highlighted the moderating role of mortality from NCDs and life expectancy. This underscores the complex and multifactorial nature of the relationship between macro-level indicators and stroke. The contrasting moderation effects observed with mortality from NCDs and life expectancy provide compelling evidence that country-level health-related indicators play a critical role in the relationship between stroke and depressive symptoms. The concepts of survivorship bias and health care infrastructure and awareness offer insights into interpreting this result. On one hand, countries with higher life expectancies and lower mortality from NCDs may boast better health care systems, potentially resulting in higher stroke incidence rates due to an aging population.³⁰ Although improved health care and early intervention in these countries may lead to higher survival rates among patients with stroke, survivors may still experience long-term consequences, including depressive symptoms. On the other hand, countries with lower mortality from NCDs may possess better health care infrastructure and awareness, leading to enhanced detection and reporting of stroke cases. Consequently, the association between stroke and depressive symptoms may become more pronounced due to increased recognition and diagnosis.

Our findings carry significant implications for both public health interventions and clinical practice. From a public health perspective, our research underscores the importance of implementing regular screening protocols for mental health concerns within the population at risk for stroke. This proactive approach can help identify individuals at risk of stroke and enable timely interventions. Additionally, our study emphasizes the critical need for accessible, responsive, and high-quality rehabilitation services for individuals affected by stroke, given the strong and enduring effects of stroke onset. Efforts to improve rehabilitation services should prioritize comprehensive support for stroke survivors, integrating rehabilitation into their long-term recovery journey.³ Furthermore, it is crucial to address the psychological needs of stroke survivors, which are often overlooked once medical services related to stroke are discontinued. Therefore, there is a pressing need for a holistic approach to stroke care, ensuring ongoing support for the psychological and physical well-being of patients with stroke.³¹

Limitations

This study had several limitations that should be acknowledged. First, our study lacks information on

stroke subtype and specific dates of stroke onset due to the nature of the database. We are unable to accurately capture the exact date of suffering a stroke for individuals, and our time units are limited to 2 to 3 years, which hinders our ability to further analyze immediate effects of stroke and differentiate the heterogeneous effects of different stroke subtypes. Second, akin to similar studies, it is important to acknowledge that the depressive symptoms and stroke reported in our study rely on self-reported responses. This is an inherent limitation of this type of investigation. Last, our sample is restricted to cases where individuals diagnosed with stroke survive until 1 wave following the diagnosis. This may lead to survivorship bias, because individuals with stroke who pass away tend to have poorer mental health. As a result, the actual effect on depressive symptoms may be larger than the current estimate.

Despite its limitations, this study boasts numerous strengths and contributes significantly to existing research. First, we conducted a robust analysis of the impact of stroke onset on depressive symptoms, spanning a substantial follow-up period and using a harmonized multicountry framework. Our findings demonstrate the consistency of these effects across diverse cultural settings and countries, offering insights into broader contextual factors and underscoring their relevance for global public health policies and practices. Second, we delved into the dynamic nature of the relationship, enabling a prospective assessment of depressive symptoms preceding stroke onset. This stands out from many others in the field of stroke research due to its distinct research design. Unlike many studies that rely solely on retrospective measures or lack prestroke assessments, our study incorporates a comprehensive analysis. Last, through meticulous empirical strategies and multiple robustness checks, we offered plausible evidence supporting a causal interpretation of our findings that enhanced the statistical conclusion. However, we acknowledge the inherent limitations of observational studies and the absence of natural experiments in our research. Therefore, we encourage future investigations to explore this relationship further in real-world contexts, using behavioral measures, experimental designs, and an expanded range of control variables to facilitate causal inferences.

CONCLUSIONS

Our research reveals a clear and full picture of stroke and depressive symptoms, highlighting a nuanced and dynamic relationship. Importantly, we identified the potential moderating influence of country-level health care service quality. We call for future investigations to delve deeper into how stroke onset impacts mental health, exploring both social psychology perspectives

and underlying pathophysiological mechanisms. Identifying different subtypes of stroke and their patterns with mental health will also help us gain a deeper understanding of this relationship.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Data S1

Tables S1–S6

Figures S1–S3

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